



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Italgapura, Rajankunte, Yelahanka, Bengaluru – 560064



IMPLEMENTATION OF AN ARTIFICIAL INTELLIGENCE DRIVEN SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION MITIGATION

A PROJECT REPORT

Submitted by,

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

Certified that this report "Implementation of an Artificial Intelligence Driven Smart Traffic Management System for Urban Congestion Mitigation" is a Bonafide work of **HEMANTH G (20221CSE0075)**, **HARSHA VARDHANA T (20221CSE0224)**, who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE AND ENGINEERING** during 2025-26.

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING DECLARATION

We the students of final year B.Tech in COMPUTER SCIENCE AND ENGINEERING at Presidency University, Bengaluru, named **Hemanth G, Harsha Vardhana T** hereby declare that the project work titled “**Implementation of an Artificial Intelligence Driven Smart Traffic Management System for Urban Congestion Mitigation**” has been independently carried out by us and submitted in partial fulfilment for the award of the degree of B.Tech in COMPUTER SCIENCE & ENGINEERING during the academic year **2025–26**. Further, the matter embodied in this project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

This project presents a Smart Traffic Management System for Urban Congestion, an AI-driven solution designed to overcome the limitations of traditional fixed-time traffic signal systems. The system leverages advanced Computer Vision and Deep Learning techniques, specifically the YOLOv8 (You Only Look Once version 8) object detection model, to identify and classify vehicles such as cars, motorcycles, buses, and trucks from real-time video feeds at road intersections. Lane-wise traffic density is computed dynamically, enabling adaptive signal control that allocates green signal time based on real-time congestion levels. The system also integrates intelligent features including emergency vehicle detection for prioritizing ambulances and critical vehicles, and an accident detection module that analyzes vehicle movement patterns and spatial overlaps to identify potential collisions. A web-based dashboard developed using the Flask framework provides real-time visualization of traffic conditions, including vehicle counts, congestion levels, emergency alerts, and accident indicators.

Experimental evaluation using simulated traffic scenarios demonstrates that the proposed system effectively detects vehicles in real time and dynamically adjusts signal timings to improve traffic flow efficiency. The system contributes to reducing congestion, travel delays, fuel consumption, and CO₂ emissions compared to conventional traffic management approaches. Additionally, it provides analytical insights such as congestion trends, accident-prone zones, and estimated vehicle delays, supporting data-driven decision-making for urban traffic planning. The implementation highlights the scalability and effectiveness of AI-based traffic monitoring systems as a cost-efficient alternative to traditional sensor-based solutions, contributing to the development of smarter and safer urban transportation systems.